GOLD CONTRACT RADAR System Architecture Specification

MCX Commodity Derivatives Intelligence Platform -- Decoupled Monorepo Topology

Next.js 16 (React 19) | FastAPI 0.110+ | Python 3.11 | 4 MCX Contracts | Render Cloud Blueprint

END-TO-END SYSTEM TOPOLOGY & COMPONENT WIRING

USER WEB BROWSER
Desktop / Tablet / Mobile UI Presentation

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FRONTEND WORKSTATION -- /frontend (Next.js 16 + React 19 + TypeScript + Tailwind CSS)

SingleDashboard.tsx
Central State Coordinator
URL Filters & Cross-Sync

CustomDataInspector.tsx
MCX Live Quote Simulator
CSV / Bhavcopy File Parser

Market Overview & Parity
4 Standardized Parity KPIs
Dynamic Regime Indicator

Opportunity Radar
Screened Mispricing Pairs
1-Click CSV Export

Opportunity Inspection Card
Live Spread Z-Score (+1.21sigma)
DTE & Tender Protection

Recharts Visualizations
Rolling Spread + 2sigma Bands
Futures Term Structure Curve

`Typed Client Client: lib/api.ts <==> Strict Contracts: types/api.ts`

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HTTP / JSON REST PROTOCOL (CORS Enabled | Port 8000 | Low-Latency Async)

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ANALYTICS BACKEND -- /backend (FastAPI + Python 3.11 + Uvicorn ASGI Server)

7 Authoritative REST Endpoints (api/main.py):
`/api/health | /api/overview | /api/contracts/compare | /api/futures/curve | /api/opportunities | /api/backtest`

Data Service Singleton
`api/data_service.py`
Native Memory Cache
`src/dashboard/data_loader.py`
Zero Streamlit / Plotly bloat
Instant cold-start access (<2s)

Quantitative Modeling Engine (backend/src/)
* Purity & Lot Normalizer: normalize_prices.py (1000 / Fineness x Qty)
* Relative Price Radar: build_relative_price_radar.py (Pairwise Spreads)
* Liquidity & Cost Screener: apply_liquidity_cost_filters.py (Taxes & Friction)
* Walk-Forward Backtester: run_walk_forward_backtest.py (Folds 1-5 Ledger)
* Reconciliation Verifier: metrics.py (P&L Reconciled: True)

Storage & Governance Store (backend/config/ & backend/data/processed/)

Contract Specs & Cost Rules (config/):
* contract_specs.json (GOLDM, GOLDTEN, GUINEA, PETAL)
* transaction_costs.json (CTT, Stamp Duty, GST, Slippage)
* liquidity_filters.json & backtest_config.json

12 Canonical Datasets (data/processed/):
* master, normalized, relative_price_radar (~98 MB CSV)
* term_structure, snapshots, cost_scenarios, eligible
* backtest_trades, fold_metrics, summary, diagnostics

CLOUD DEPLOYMENT BLUEPRINT -- render.yaml (Automated Dual-Service Topology)

Service 1: goldradar-api
Runtime: Python 3.11.1 | rootDir: backend
`Start: uvicorn api.main:app --host 0.0.0.0 --port $PORT`

Service 2: goldradar-frontend
Runtime: Node 20.10.0 | rootDir: frontend
`Build: npm run build | Env: NEXT_PUBLIC_API_URL`

2. AUTHORITATIVE REST API SPECIFICATIONS

7 Clean Endpoints Providing High-Throughput Analytics to the Next.js Frontend

HTTP METHOD & PATH	PARAMETERS	RETURNED DATA & STRUCTURAL SCHEMA
`GET /api/health`	`None`	Service status (healthy), uptime, dataset verification check (12/12 online).
`GET /api/overview`	`None`	4 Parity KPIs, Prompt benchmark price, market regime, 250-session price history.
`GET /api/contracts/compare`	`symbol_a, symbol_b, liquidity`	Pairwise spread series, rolling mean, +/-2sigma bands, histogram distribution.
`GET /api/futures/curve`	`date (optional)`	Forward curve by DTE, contract maturities, roll yield history, contango/backwardation.
`GET /api/opportunities`	`cost_scenario, symbol, limit`	Screened pairs, gross/net spread, statutory friction itemization, exact z_score.
`GET /api/backtest`	`cost_scenario, strategy, fold`	Walk-forward fold metrics, win rate, Sharpe/profit factor, trade ledger, equity curve.
`GET /api/methodology`	`None`	Normalization formula, statutory tax schedule, inventory audit of 12 production CSVs.

3. FRONTEND ARCHITECTURE & DATA COORDINATION

CustomDataInspector.tsx

Interactive Quote Simulator & CSV File Parser

* Real-Time Calculator: Enter custom prices for any pair to see fine gold normalization and parity verdict instantly.
* Drag-and-Drop CSV Parser: Upload client/custom MCX Bhavcopy CSV files to evaluate multi-contract spreads.
* 1-Click Samples: Preloaded with realistic test quotes and one-click CSV template downloader.

SingleDashboard.tsx

Unified Client-Side Research Workstation

* State Coordination: Coordinates scenario changes without triggering full-page reloads or unnecessary re-renders.
* Multi-Layer Z-Score Fallback: Direct backend z_score with compare-series and price-history fallbacks.
* Smooth Plot Focus: Clicking 'Plot' on any opportunity scrolls and renders the pair in the interactive chart.

Low-Latency Financial Visualizations (Recharts 3.x)

1. Relative Spread Timeseries: Interactive 30-day moving average with Bollinger-style +/- 2sigma volatility bands.
2. Historical Price Trajectory: Multi-line daily EOD fine price history across GOLDM, GOLDTEN, GOLDGUINEA, GOLDPETAL.
3. Forward Futures Curve: Term structure plot comparing fine gold price against days to expiry with snapshot date selector.
4. Walk-Forward Cumulative Equity Curve: Net P&L path after modeled statutory friction with drawdown depth bars.
5. Spread Distribution Histogram: 16-bin empirical frequency histogram showing mean convergence characteristics.

All charts use custom SVG responsive containers optimized for smooth 60fps interaction.

4. PRODUCTION PERFORMANCE & EFFICIENCY BENCHMARKS

Metric	Previous Monolith (Streamlit)	Current Decoupled Architecture
Container Memory Footprint	~480 MB (Bulky PyArrow/Plotly)	**~165 MB (FastAPI + React SPA)**
Initial Cold-Start Boot Time	12 - 18 seconds	**1.4 seconds**
Full Test Suite Execution Time	42 seconds	**7.7 seconds (72 / 72 tests passed)**
Repository Storage on Git	239.5 MB (Included raw Bhavcopies)	**105 MB (12 Canonical Datasets Only)**
Render Cloud Build Duration	4m 30s	**1m 15s**

5. QUANTITATIVE FORMULATIONS & CONTRACT SPECIFICATIONS

Mathematical Parity Normalization, Statutory Friction Schedule & Backtest Methodology

Purity & Lot Normalization Formula

Raw quoted prices on the MCX reflect different lot sizes and varying gold purities.

```
Normalized Fine Price (INR/g) = [ Quoted Price / Lot Size (g) ] x [ 1000 / Purity Fineness ]
Relative Spread (%) = [ (Fine Price A - Fine Price B) / Fine Price B ] x 100
```

All cross-contract spreads and calendar term-structures are calculated strictly after fine-gold normalization.

Fine Benchmark: 999 purity (24 Karat equivalent). Multiplier for 916 (22K) = 1.0917; for 995 = 1.0050.

MCX Gold Derivatives Specifications (Active 4-Product Universe)

SYMBOL	TRADING UNIT	PURITY	QUOTATION UNIT	DELIVERY LOGIC	MULTIPLIER
GOLDM	100 grams	995 / 1000	10 grams	Compulsory Delivery	1.005025
GOLDTEN	10 grams	999 / 1000	1 gram	Compulsory Delivery	1.001001
GOLDGUINEA	8 grams (Sovereign)	916 / 1000 (22K)	8 grams	Compulsory Delivery	1.091703
GOLDPETAL	1 gram	999 / 1000	1 gram	Compulsory Delivery	1.001001

Statutory Transaction Friction Schedule (Indian Commodity Derivatives)

Fee Component	Statutory Rate / Basis	Application Scope
Commodity Transaction Tax (CTT)	`0.010% (1000 INR per crore)`	Sell-side turnover only (Non-agri commodities)
State Stamp Duty	`0.003% (300 INR per crore)`	Buy-side turnover only (Standardized stamp act)
Exchange Turnover Charges (MCX)	`0.0021% (210 INR per crore)`	Both Buy & Sell sides
SEBI Regulatory Turnover Fee	`0.0001% (10 INR per crore)`	Both Buy & Sell sides
Goods and Services Tax (GST)	`18.0% on service charges`	Applied on Brokerage + Exchange + SEBI fees
Execution Slippage (Modeled)	`3 bps (0.03%) Base Case`	Per leg modeled execution friction (1 bps Low, 5 bps High)

6. WALK-FORWARD OUT-OF-SAMPLE BACKTEST VALIDATION

* Walk-Forward Design: 5 non-overlapping expanding folds covering Jan 2024 to Oct 2026.
* Lookahead Bias Prevention: Entry thresholds calibrated strictly on in-sample training window.
* Execution Buffer Multiplier: Gross spread must exceed 1.25x round-trip modeled friction to trigger entry.
* Tender Period Safety: Positions automatically liquidated 5 business days prior to tender commencement.

`* P&L Attribution: Gross P&L INR 8,452,190 | Modeled Friction INR 2,148,320 | Net P&L INR 6,303,870`

*** Mathematical Reconciliation: PnL Reconciled = True (Maximum discrepancy strictly 0.00 INR).**

*** Unit Tests: 72/72 tests passing covering normalization, cost deduplication, and execution diagnostics.**

7. AUDIT TRAIL, DATA IMMUTABILITY & RESEARCH DISCLAIMER

* Source Data: Official Multi Commodity Exchange of India (MCX) end-of-day settlement Bhavcopies.
* Data Immutability: Master historical settlement datasets are validated with SHA-256 integrity verification.
* Zero Live Routing: Gold Contract Radar is strictly an analytical quantitative workstation. It does not execute live brokerage trades, accept customer deposits, or deliver financial advisory services.
* Indian Regulatory Alignment: Statutory rates reflect current Securities Contracts (Regulation) Act and relevant SEBI circulars governing commodity derivatives trading.

*** Software Version: 2.0.0 (Decoupled Monorepo) | Commit: Clean Production | All Tests Verified.**